



TITLE PAGE

- Problem Statement ID – 26149
- Problem Statement Title - Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization
- Theme - Blockchain& Cybersecurity
- PS Category- Software
- Team ID - 165039
- Team Name (Registered on portal): DataSentinels



THE PROBLEM

Deleted data may still be recoverable.

Traditional data deletion does not always guarantee that information is permanently removed. At the same time, deleted digital evidence can be difficult to recover and verify.

OUR PROPOSED SOLUTION

DATA RECOVERY

Recover deleted files through file carving and reconstruction techniques.

SECURE ERASURE

Securely sanitize data and verify the sanitization process

VERIFIED REPORTING

Generate traceable reports to document recovery and erasure results.

KEY DIFFERENTIATOR

Recovery + Sanitization + Verification in one integrated platform.

RECOVERY

Recover deleted files.



JPG



DOCX



ZIP



PDF

Recover

Analyze

SECURE SANITIZATION

Securely erase files, folders & controlled disk images



File



Folder



Disk

ANALYSIS & SCORING

Validate files & generate confidence scores



Signature



Structure



Confidence

Verify

Sanitize

VERIFY & REPORT

Compare pre/post recovery.
Verify PASS/FAIL. Generate report.



Before/
After



Pass/Fail



Audit Log



Report

FEASIBILITY & VIABILITY



TECHNICAL FOUNDATION

- Sleuth Kit
- Raw File Carving
- Validation



CONTROLLED TESTING

- Controlled Environment
- MVP is developed and tested safely using controlled disk images



RECOVERY & VERIFICATION

- Filesystem recovery
- Fragment reconstruction
- File validation & confidence scoring

Technical Foundation

Uses established forensic techniques.

Practical Requirements

Addresses real-world forensic recovery needs.

Feasibility

Proven forensic methods
Sleuth Kit, signature-based carving, validation.

Viability

Controlled and verifiable results
Validation, pre/post comparison.

Forensic Success

Evidence-driven recovery & verifiable sanitization

IMPACT

-  **Verified Results**
-  **Secure Data Erasure**
-  **Efficient Recovery & Analysis**



BENEFITS

-  **Protect Sensitive Data**
-  **Reduce Data Exposure Risks**
-  **Recover Deleted Evidence**
-  **Verified Recovery & Erasure**

ACADEMIC RESEARCH

- **Forensic Science International: Digital Investigation (2023)** : Identification of data wiping tools based on deletion patterns in ReFS \$Logfile — research on data wiping tools, deletion patterns, and forensic detection.
- **arXiv Research (2026)** : Byte-level generative predictions for forensics multimedia carving — explores generative models for fragmented file recovery and byte-level fragment prediction.
- **CUSTOS** : Toward Forensic-Ready Zero Trust at the Capture-Containment Boundary.

TECHNICAL RESOURCES

- The Sleuth Kit (TSK) : Sleuth kit Documentation and developers guide.

